



# CCS3356

## Natural Language Processing

### NLP Assignment – Project Validation Submission

#### Group Information

Group Name:

- **Logic Legends** Group Number:
- **12**

Project Title:

- **Sentiment Analysis of Customer Reviews Using Machine Learning and Deep Learning** Group Members:

|   | Student ID     | Student Name |
|---|----------------|--------------|
| 1 | CIT-24-01-0241 | B Premnath   |
| 2 | CIT-24-01-0010 | M S M Simak  |
| 3 | CIT-24-01-0298 | A S M AASIM  |

#### ➤ Section 1 – Project Overview

1. What NLP problem/application are you attempting to solve?

- The project revolves around the problem of performing Sentiment Analysis on Customer Reviews using NLP-based solutions. Sentiment analysis refers to the ability of automated systems to identify and categorize opinions expressed in texts to predefined sentiment categories such as positive, negative, or neutral. Texts that are used for sentiment analysis typically include customer reviews about their experiences with services and products offered by companies. These texts are rich in content regarding users' satisfaction, quality of products, performance of services, and other important aspects related to the customer experience. Nevertheless, manual analysis of hundreds of reviews is inefficient and does not guarantee accurate results. Thus, the goal of this project is to create an automatic system that can classify customer sentiments from texts. To accomplish this objective, NLP pre-processing techniques would be used to analyze the input data and perform the classification using machine learning and deep learning algorithms. Several different models would be created to evaluate which algorithm works better. This project offers a comparison between machine learning and deep learning algorithms in order to choose the best performing one.

## 2. Why is this problem important?

Who are the intended users and what value does your solution provide?

- Customer feedback is instrumental in enhancing products and services. Businesses get hundreds of feedbacks from their customers via e-commerce sites, social media, web sites, and mobile applications. The process of reading and analyzing all feedbacks manually may be too expensive. Thus, an automated sentiment analysis system will help business people to analyze their clients' sentiments.

The users who may use this system include:

- ✓ Owners of companies
- ✓ Managers of production
- ✓ Marketing experts
- ✓ Customer care departments
- ✓ Consumer behavior researchers

The solution will bring the following benefits to its users:

- ✓ Categorization of feedbacks into sentiment classes
- ✓ Identification of clients' opinions about products/services
- ✓ Analysis of existing issues to enhance products
- ✓ Efficient decision making based on collected data

## 3. What will be the final output of your application?

Examples:

- Classification label
- Generated summary

- Translation
- Chatbot response
- Answer to a question

Briefly explain:

Output Type: Classification

Label

Explanation:

The application will analyze a customer review entered by a user and predict its sentiment category. The output will be one of the following labels:

- Positive
- Negative
- Neutral

For example:

Input

Review:

"The product quality is excellent and delivery was very fast."

Output:

Positive

The system may also display confidence scores for each prediction and visual summaries of sentiment distributions.

## ➤ Section 2 – Dataset Information

### 4. Dataset Details

- Dataset Name:
  - ✓ Amazon Customer Reviews Dataset
- Dataset Source(URL):
  - ✓ <https://www.kaggle.com/datasets/datafiniti/consumer-reviews-of-amazonproducts>

- Number of Records:
  - ✓ Approximately 34,000 customer reviews
- Number of Classes (if applicable):
  - ✓ 3 Sentiment Classes
    - Positive
    - Neutral
    - Negative

#### 5. Why is this dataset suitable for your project?

- This dataset is highly suitable because it contains a large number of real-world customer reviews collected from actual users. The reviews represent diverse opinions regarding products and services. Since sentiment analysis relies heavily on opinion-rich textual data, customer reviews provide an ideal source of information. The dataset contains enough records to train both Machine Learning and Deep Learning models effectively. It also includes textual reviews that can be transformed into sentiment labels for supervised learning. Furthermore, the dataset reflects realistic language usage patterns, including informal expressions, abbreviations, and customer-specific terminology. These characteristics make the dataset practical for evaluating the performance of NLP models in real-world scenarios.

#### 6. What challenges exist within the dataset?

Examples:

- Missing values
- Imbalanced classes
- Noisy text
- Multiple languages
- Ethical concerns

Explain:

- Missing Values

There might be some cases where the customer reviews do not contain enough information or have missing text field data, which will need to be detected.

- Noise in Text

Spelling errors, redundant letters, extra punctuation, emojis, URL addresses, and special characters might be found in customer reviews, which could interfere with model training.

- Class Imbalance

Usually, the positive review number is much higher than the negative ones, which creates a risk of a skewed distribution.

- Informal Writing Style

Slang, abbreviations, colloquial expressions, and informal language usage are widespread in customers' reviews, making it difficult for a model to understand what they mean.

- Duplicate Data Points

It is possible that there will be duplicate reviews which should be taken into account during processing and learning.

- Ethical Issues

Customer reviews might include personal information, profanity, and biases, therefore, care should be taken regarding data protection.

### ➤ Section 3 – Individual NLP Pipeline

**Each member must complete this section separately.**

#### **Member 01 ( Premnath )**

List the steps of NLP pipeline you plan to perform and justify each.

| Step            | Justification                                                                                                                                                                                                                                                                                      |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Collection | The first step is to gather customer reviews from a reliable dataset. These reviews provide the information needed to understand customer opinions and sentiments. A diverse dataset helps the model learn from different types of feedback and improves its ability to make accurate predictions. |

|                           |                                                                                                                                                                                                                                                                          |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Cleaning             | Raw text data often contains errors, duplicate entries, missing values, links, and unwanted symbols. Cleaning the data removes these unnecessary elements and ensures that the reviews are clear and consistent, which improves the overall quality of the analysis.     |
| Tokenization              | Tokenization breaks each review into smaller units called tokens, usually words. This makes it easier for the computer to process and analyze the text because it can work with individual words rather than entire sentences.                                           |
| Stop Word Removal         | Many common words such as "the," "is," and "and" appear frequently in text but do not contribute much to the sentiment of a review. Removing these words helps reduce noise and allows the model to focus on more meaningful terms.                                      |
| Lemmatization             | Words can appear in different forms while having the same meaning. Lemmatization converts words into their base form, such as changing "running" to "run." This helps reduce the number of unique words and improves consistency in the dataset.                         |
| TF-IDF Feature Extraction | Machine learning models cannot directly understand text. TF-IDF converts text into numerical values by measuring how important a word is within a review compared to the entire dataset. This helps the model identify the most relevant words for sentiment prediction. |
| Model Training            | During this stage, machine learning and deep learning algorithms are trained using the processed data. The models learn patterns and relationships between words and sentiments, allowing them to classify reviews as positive, negative, or ne                          |

## ML Model Selected

Why was it selected?

### ☐ Random Forest Classifier

Random Forest is robust, handles large datasets effectively, reduces overfitting through ensemble learning, and performs well on text classification tasks.

## DL Model Selected

Why was it selected?

### ☐ LSTM (Long Short-Term Memory)

LSTM networks are specifically designed for sequence data and can capture contextual relationships within text, making them highly effective for sentiment analysis.

What contribution are you responsible for in the project?

- Data preprocessing

- Feature extraction
- Random Forest implementation
- Performance evaluation
- Documentation support

## Member 02 ( Simak )

List the steps of NLP pipeline you plan to perform and justify each.

| Step                 | Justification                                                                                                          |
|----------------------|------------------------------------------------------------------------------------------------------------------------|
| Data Cleaning        | Removes noise, missing values, special characters, and irrelevant information to improve the quality of the text data. |
| Lowercase Conversion | Converts all text to lowercase to ensure consistency and prevent duplicate representations of the same word.           |
| Tokenization         | Splits text into individual words or tokens, making it easier for the model to process and analyze.                    |
| Stop Word Removal    | Eliminates common words that contribute little meaning, helping reduce data dimensionality and improve efficiency.     |
| Stemming             | Reduces words to their root forms, allowing similar words to be treated as a single feature.                           |
| Vectorization        | Transforms textual data into numerical representations that machine learning algorithms can understand.                |
| Model Training       | Learns patterns and relationships from the processed text data to perform sentiment classification.                    |
| Evaluation           | Measures and compares model performance using metrics such as accuracy, precision, recall, and F1-score.               |

## ML Model Selected

Why was it selected?

### □ Support Vector Machine (SVM)

SVM performs exceptionally well in high-dimensional text classification problems and has proven effectiveness in sentiment analysis applications.

## DL Model Selected

Why was it selected?

### • GRU (Gated Recurrent Unit)

GRU offers performance similar to LSTM while requiring fewer parameters and shorter training times.

What contribution are you responsible for in the project?

- Data cleaning
- SVM implementation
- GRU model development
- Experimental analysis

- Report writing

### Member 03 ( Aasim )

List the steps of NLP pipeline you plan to perform and justify each.

| Step                | Justification                                                                                                                                                                                                                         |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dataset Exploration | Analyze the dataset to understand its size, structure, class distribution, and data quality. This helps identify potential issues such as missing values, imbalanced classes, or duplicate records before preprocessing and modeling. |
| Text Cleaning       | Remove irrelevant information such as special characters, punctuation, URLs, numbers, and extra spaces to improve text quality and ensure consistent input for NLP models.                                                            |
| Tokenization        | Split text into smaller units called tokens (words or subwords). Tokenization is essential for transforming raw text into a format that can be processed by machine learning and deep learning algorithms.                            |
| Lemmatization       | Reduce words to their base or dictionary forms while preserving their meaning. This helps group similar words together and reduces vocabulary size, improving model performance.                                                      |
| Word Embedding      | Convert text into dense numerical vectors that capture semantic and contextual relationships between words. Word embeddings enable models to better understand the meaning of text data.                                              |
| Model Training      | Train machine learning and deep learning models using the processed text data to learn patterns and relationships that can accurately predict sentiment labels.                                                                       |
| Evaluation          | Assess the effectiveness of the trained models using metrics such as accuracy, precision, recall, and F1-score. This helps determine how well the models perform on unseen data.                                                      |
| Visualization       | Present data distributions, model performance, and analysis results through charts and graphs. Visualizations make findi                                                                                                              |

### ML Model Selected

Why was it selected?

#### ☐ Naive Bayes

Naive Bayes is computationally efficient, simple to implement, and widely used as a baseline model for text classification tasks.

### DL Model Selected

Why was it selected?

#### • RNN for Text Classification

RNNs can identify important local patterns and phrases in text, making them effective for sentiment classification.



What contribution are you responsible for in the project?

- Dataset exploration
- Naive Bayes implementation
- RNN development
- Result visualization
- Presentation preparation

#### ➤ Section 4 – Model Comparison Plan

7. Summarize all selected models.

| Member   | Machine Learning (ML) Model  | Deep Learning (DL) Model           |
|----------|------------------------------|------------------------------------|
| Member 1 | Random Forest                | LSTM (Long Short-Term Memory)      |
| Member 2 | Support Vector Machine (SVM) | GRU (Gated Recurrent Unit)         |
| Member 3 | Naive Bayes                  | RNN (Convolutional Neural Network) |

8. What evaluation metrics will be used to select the best ML and DL models?

Examples:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

Provide justification.

- Accuracy

Measures overall correctness of predictions.

- Precision

Measures the proportion of correctly predicted positive instances among all positive predictions.

- Recall

Measures the ability to identify actual positive instances.

- F1 Score

Balances Precision and Recall, especially useful when class distributions are imbalanced.

- Confusion Matrix

Provides detailed information regarding correct and incorrect classifications.

- ROC-AUC

Measures the ability of models to distinguish between sentiment classes.

- Justification

Using multiple evaluation metrics provides a comprehensive assessment of model performance. Accuracy alone may not adequately represent effectiveness when class imbalance exists. Therefore, Precision, Recall, F1 Score, and ROC-AUC will be used together to select the best-performing models.



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## Section 5 Final Application Plan

9. What type of application will be developed?

- Web Application
- Desktop Application
- API
- Mobile Application
- CLI Application

### ➤ Web Application

10. Briefly describe how the application will work.

Provide a simple workflow. (simple system diagram)

#### Workflow

User Inputs Customer Review



Text Preprocessing



Feature Extraction



ML/DL Model Prediction

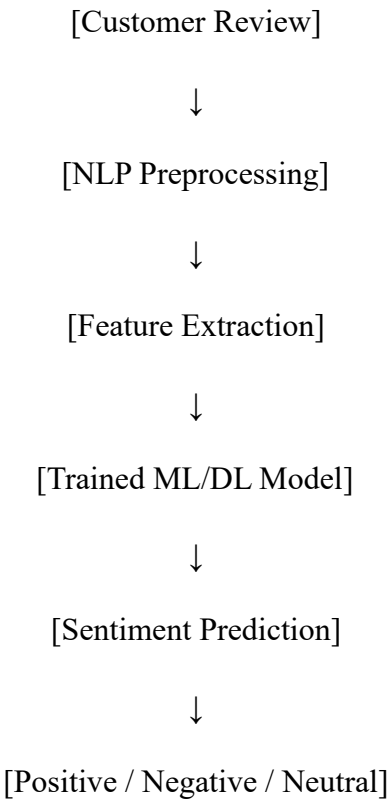


Sentiment Classification



Display Result to User

**Simple System Diagram**



## Section 6 Workload & Git Management

### 11. Work Distribution

| Task                     | Responsible Member(s) |
|--------------------------|-----------------------|
| Dataset Identification   | All Members           |
| Data Preprocessing       | Premnath, Simak       |
| Feature Engineering      | Premnath              |
| Model Development        | All Members           |
| Model Evaluation         | All Members           |
| Application Development  | Premnath, Aasim       |
| Report Writing           | Simak                 |
| Presentation Preparation | Aasim, Premnath       |

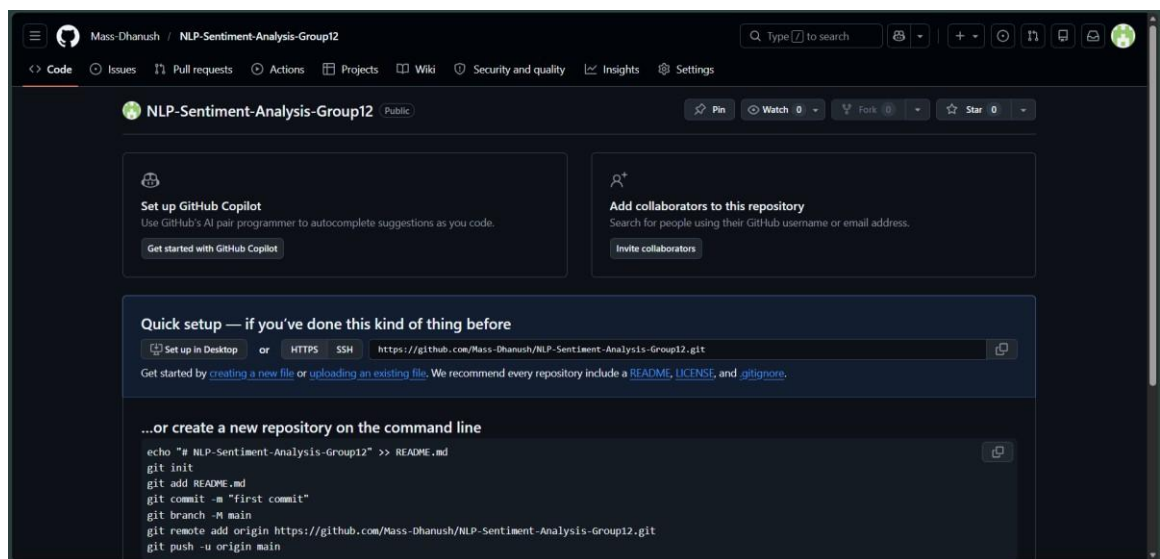
### 12. Git Repository Provide:

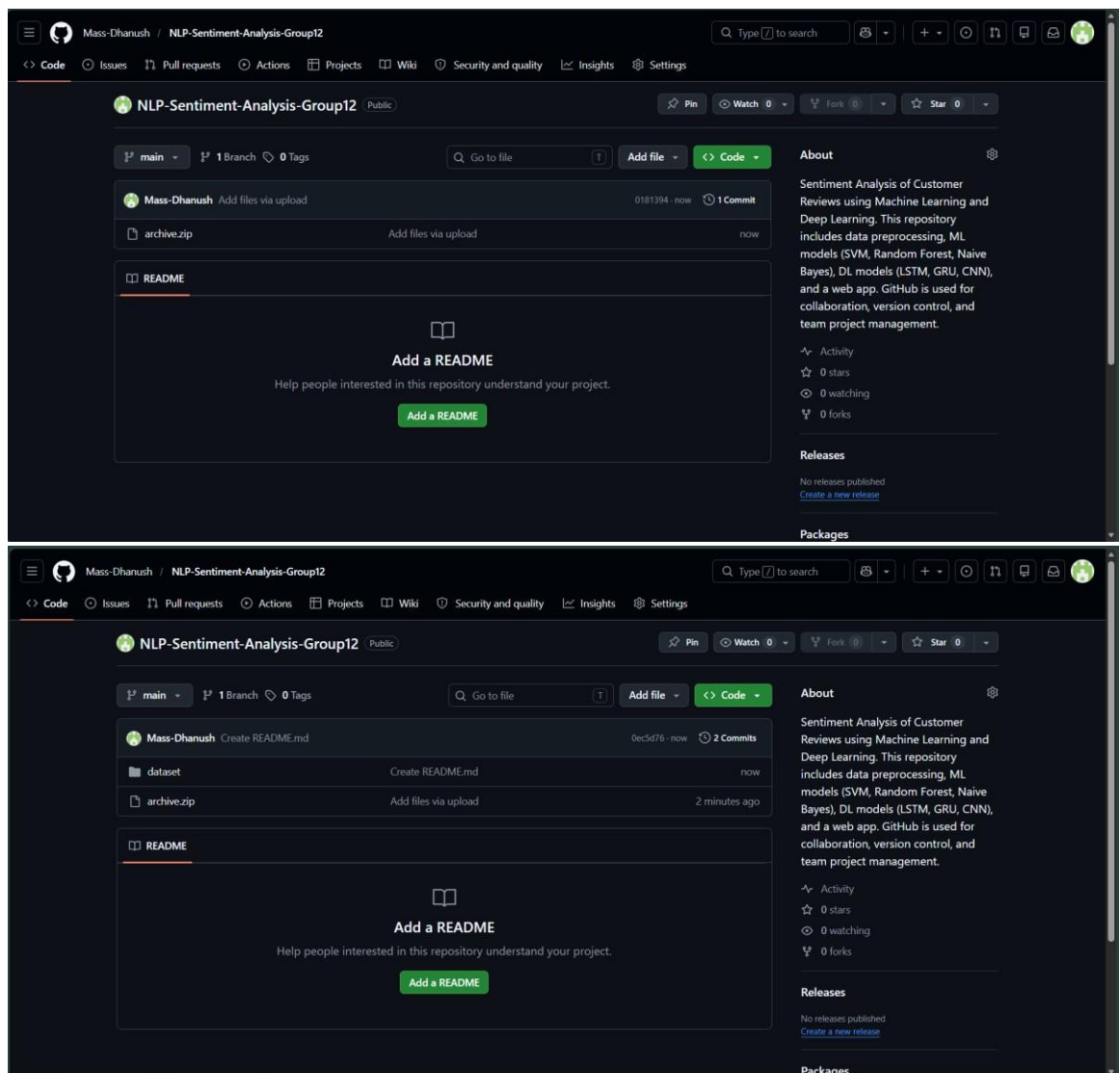
- Repository Link:

[Mass-Dhanush/NLP-Sentiment-Analysis-Group12](https://github.com/Mass-Dhanush/NLP-Sentiment-Analysis-Group12) at Premnath\_branch

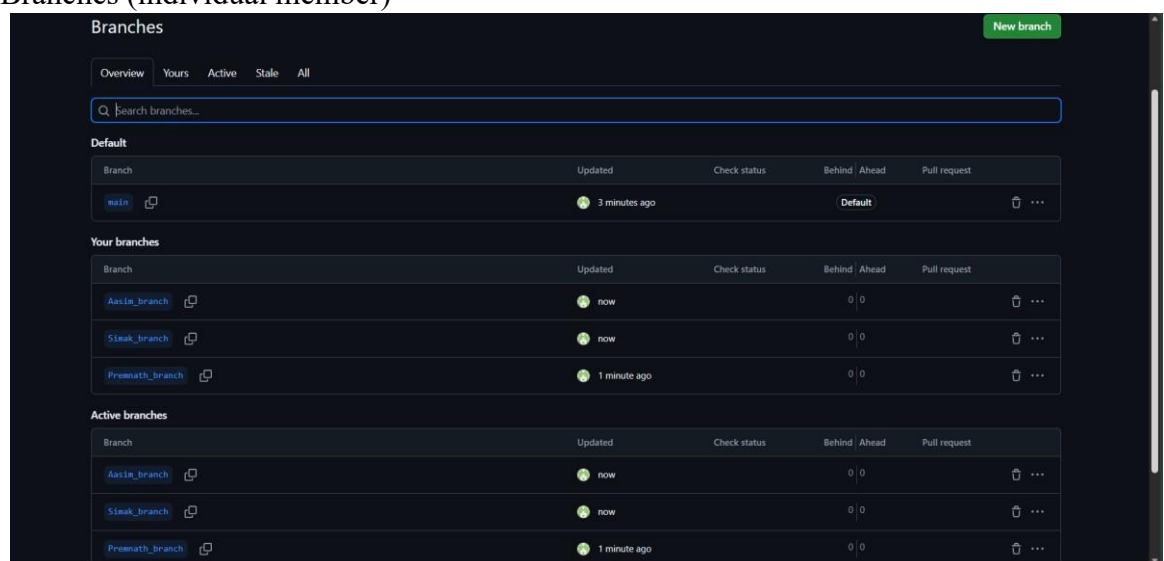
Attach screenshots of

- Repository structure





- Branches (individual member)



## Section 7 Ethics & Responsible AI

### 13. Identify potential biases and Ethical problems in the project.

- It is also possible that there is bias in the dataset with customer reviews that do not represent different user populations equally. This could create predictions which are skewed because some products and demographics could be over-represented. Reviewers might also use offensive language or make subjective comments. There could also be issues of privacy for customers.

### 14. Could your system produce harmful or misleading outputs?

Provide examples.

Yes.

Examples include:

- Misclassifying negative reviews as positive.
- Misinterpreting sarcasm or irony.
- Incorrectly identifying mixed emotions.
- Producing biased predictions due to imbalanced training data.

For example:

Review:

"Great product, but it stopped working after one week."

The system may incorrectly classify it as Positive due to the presence of the word "Great."

### 15. How will you reduce these risks you identified in Questions 13 and 14?

Several strategies will be employed:

- Dataset balancing techniques.
- Data cleaning and quality assurance.
- Cross-validation during training.
- Multiple evaluation metrics.
- Error analysis on misclassified samples.
- Continuous testing and model improvement.
- Removal of sensitive information where applicable.

These measures will help improve fairness, reliability, and overall model performance.

## 16. What limitations should future users be aware of?

The system may struggle with:

- Sarcasm and irony.
- Context-dependent meanings.
- Multilingual reviews.
- Extremely short reviews.
- Highly domain-specific terminology.
- New vocabulary not present during training.

Users should understand that sentiment predictions are probabilistic estimates rather than absolute judgments and should be used as decision-support tools rather than sole decisionmaking mechanisms.

### ➤ Declaration

We confirm that the information provided accurately reflects the current status and planned direction of the project.

Member 1 Signature: \_\_B Premnath\_\_\_\_\_

Member 2 Signature: \_\_\_\_\_M S M Simkak\_\_\_\_\_

Member 3 Signature: \_\_\_\_\_A S M Aasim

Date: \_\_\_\_\_16/06/2026\_\_\_\_\_